

1

Input			Output		
0	0	0	1	1	1
0	0	1	1	1	0
0	1	0	1	0	1
0	1	1	1	0	0
1	0	0	0	1	1
1	0	1	0	1	0
1	1	0	0	0	1
1	1	1	0	0	0

$A_0 \leftrightarrow B_7$

$A_1 \leftrightarrow B_6$

$A_2 \leftrightarrow B_5$

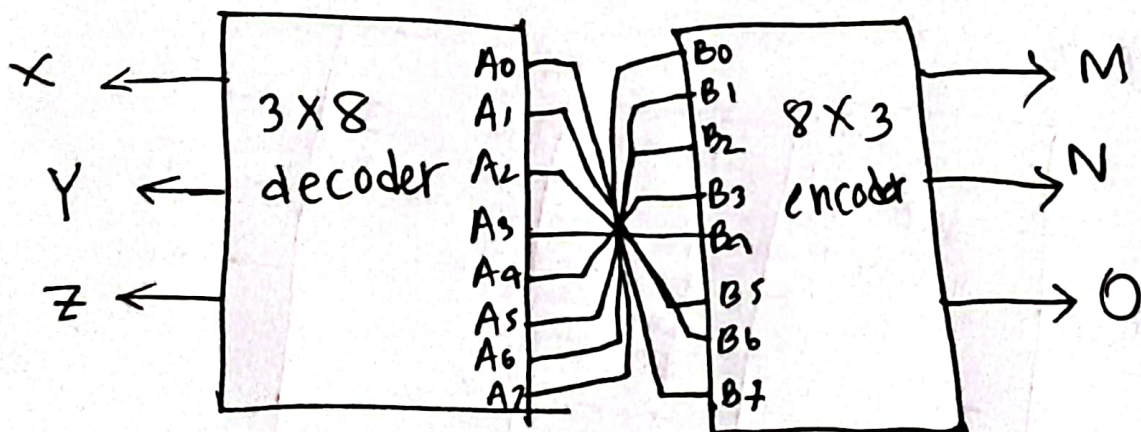
$A_3 \leftrightarrow B_4$

$A_4 \leftrightarrow B_3$

$A_5 \leftrightarrow B_2$

$A_6 \leftrightarrow B_1$

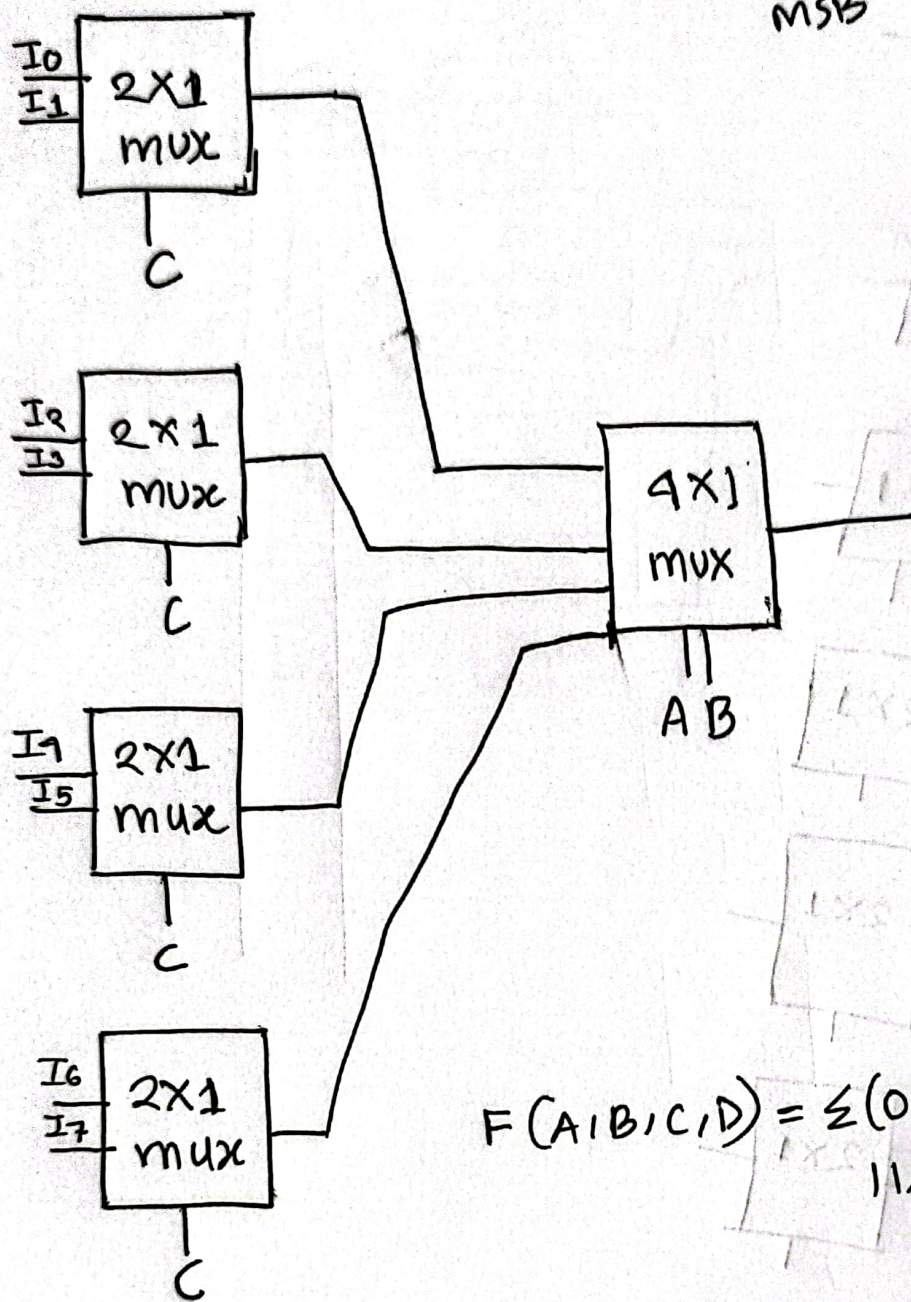
$A_7 \leftrightarrow B_0$



Design full adder using 3

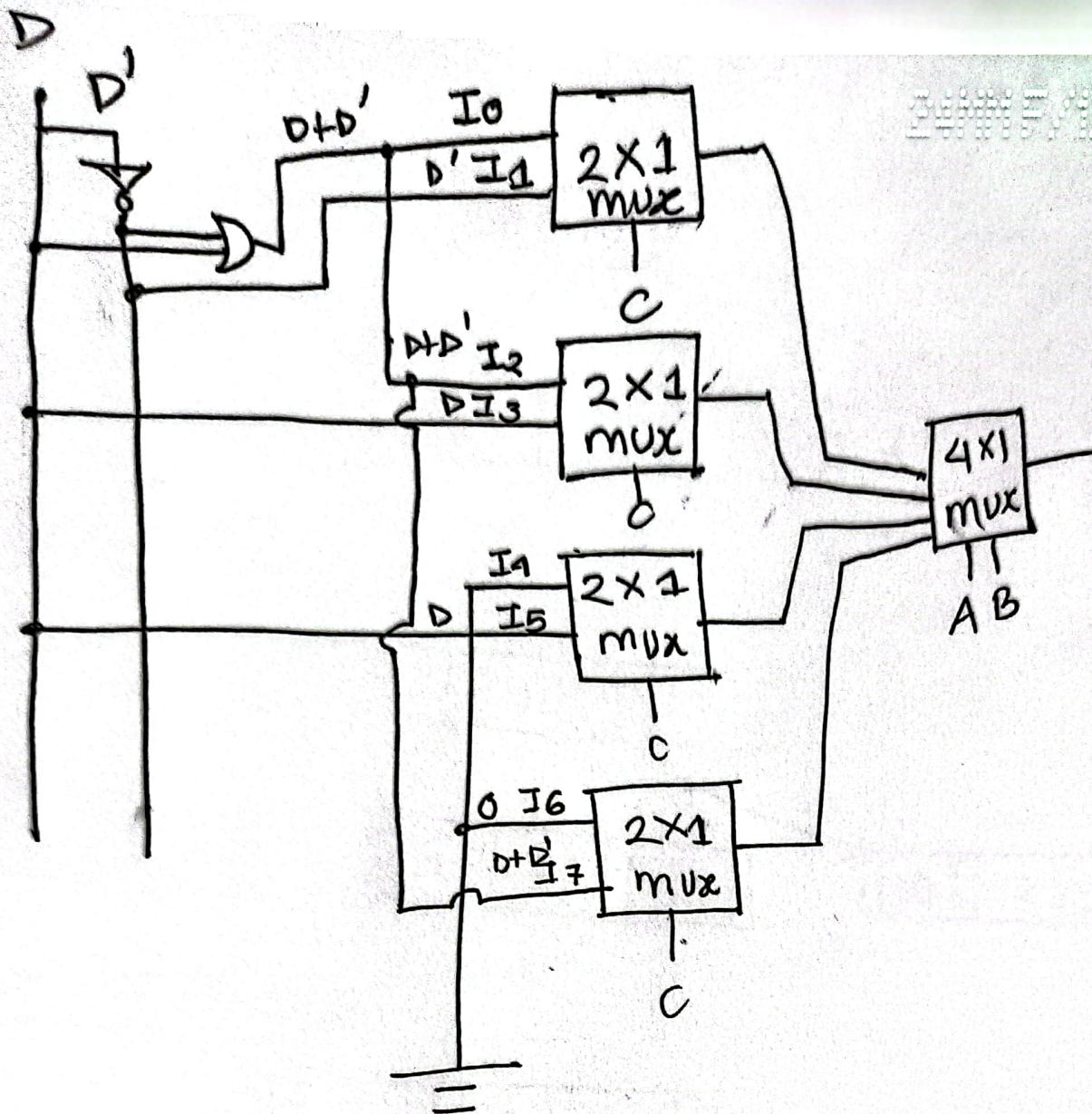
2

MSB ← ABCD → LSB



$$F(A, B, C, D) = \sum(0, 1, 2, 7, 8, 10, 11, 13, 15)$$

	I ₀	I ₁	I ₂	I ₃	I ₄	I ₅	I ₆	I ₇
D'	0	1	2	3	4	5	6	7
D	8	9	10	11	12	13	14	15
	1	D'	1	D	0	D	0	1



3

$$x \geq 8 \Rightarrow x_3 x_2 x_1 x_0 : x_3 = 1 \dots (1)$$

4 bit Y divisible by 12 = 0000, 1100

$$Y_3 Y_2 Y_1 Y_0 : Y_1, Y_0 = 0, Y_3 = Y_2 = 0/1$$

$$C = x_3 [(Y_3 \oplus Y_2)' \cdot (Y_0 \cdot Y_1)']$$

$C = 1$ addition $s = 0$ required & vice versa. 4 bit parallel adder C_0 line must receive.

$$[x_3 [(Y_3 \oplus Y_2)'] \cdot (Y_0 \cdot Y_1)']'$$

